

Biodiversity Notes

Section 1: What is Biodiversity?

A World Rich in Biodiversity

- Bio diversity short for biological diversity, is:
 - the variety of organisms in a given area
 - the genetic variation within a population
 - the variety of species in a community
 - the variety of ecosystems in an ecosystem.
- Certain areas of the planet, such as tropical rain forests, contain an extraordinary variety of species.

Unknown Diversity

- The number of species known to science is about 1.7 million, most of which are 8.7 million.
 - Actual number of species on Earth is identified and described
- New species are considered known when they are classified & described scientifically.

Levels of Diversity

- Biodiversity can be studied and described at three levels:
 - species diversity
 - ecosystem diversity
 - intermediate
- Species diversity refers to all the variety between populations of species, as well as between different species.
- Ecosystem diversity refers to the genetic variations of habitats, communities, and ecological processes within and between ecosystems.
- Genetic diversity refers to all the genetic variations contained within all members of a population.
- A gene is a segment of DNA that is located in a chromosome and that code for a specific hereditary trait.

Benefits of Biodiversity

- Biodiversity can affect the stability of ecosystems and the health of populations.

Species Are Connected to Ecosystems

- Every species is probably either dependent upon by at least one other species in ways that are not always obvious.
- A Keystone species is a species that is critical to the functioning of the ecosystem in which it lives because it affects the survival and abundance of many other species in its community.
- An example is the sea otter.
- genetic diversity increases the chances that some members of the population may survive environmental pressures or changes.
- species with low genetic diversity are less likely to survive such pressures.

Species and Population Survival

- When a population shrinks, its genetic diversity decreases as though it is passing through a bottleneck.
- The members of the population may then become more likely to inherit genetic diseases.

Medical and Industrial Uses

- About 25% of the drugs prescribed in the United States are derived from plants, and almost all of the antibiotics are derived from chemicals found in Fungi.

Agricultural Uses

- Most of the crops produced around the world originated from a few areas of high biodiversity.
- Most new crop varieties are hybrids, or crops developed by combining genetic material from other populations.

Ethics, Aesthetics, and Recreation

- ecotourism is a form of tourism that supports the conservation and sustainable development of ecologically unique areas.

Biodiversity at Risk

- The extinction of many species in a relatively short period of time is called a mass extinction.

Current Extinctions

- The rate of extinctions is estimated to have increased by a multiple of 1000 since 1800, with up to 25 percent of all species on Earth becoming extinct between 1800 and 2100.
- The current mass extinction is different from those of the past because humans are the primary cause of the extinctions.

Species Prone to Extinction

- Large populations that adapt easily to many habitats are less likely to become extinct.
- However, small populations in limited areas can easily become extinct.
- Species that are especially at risk of extinction are those that need specific habitat, those that need large or special habitats, and those that are exploited by humans.
- An endangered species is a species that has been identified to be in danger of extinction throughout all or a significant part of its range, and that is thus under protection by regulations or conservation measures.
- A threatened species is a species that has been identified to be likely to become endangered in the foreseeable future.

How Do Humans Cause Extinctions?

- The major causes of extinction today are
 - habitat destruction
 - invasive species
 - pollution
 - overharvesting

Habitat Destruction and Fragmentation

- In the process, we alter or destroy the habitats of other species.
- It is estimated that habitat loss causes almost 75% of the extinctions now occurring.

Invasive Exotic Species

- An invasive species is a species that is not native to a particular region.
- Exotic species can harm or kill native species that have no natural defenses against them.

- Poaching is the illegal harvesting of fish, game, or other species.

Pollution

- The bald eagle was endangered because of a pesticide known as DDT. Although DDT is now illegal to use in the United States, it is still manufactured here and used around the world.

Areas of Critical Biodiversity

- An endemic species is a species that is native to a particular place and that is found only there.
- Ecologists often use the numbers of endemic species of plants as an indicator of overall biodiversity because plants form the basis of ecosystems on land.

Tropical Rain Forests

- Biologists estimate that over half of the world's species live in these forests even though they cover only 2% of the Earth's land surface.

Coral Reefs and Coastal Ecosystem

- Nearly 60% of Earth's coral reefs are threatened by human activities, such as pollution, development along waterways, and overfishing.

Islands

- When an island rises from the sea, it is colonized by a small number of species from the mainland. These colonizing species may then evolve into several new species.

Biodiversity Hotspots

- The most threatened areas of high species diversity on Earth have been labeled biodiversity hotspot and include mostly tropical rainforests, coastal areas, and islands.
- Most of these hotspots have lost at least 70% of their original natural vegetation.

Biodiversity in the United States

- The United States includes a wide variety of unique ecosystems, including the deserts, the California coastal region, Hawaii, the Midwestern prairies, and the forests of the Pacific Northwest.
- The Hawaiian islands, a biodiversity hotspot, is home to 3,488 native plant species.

Saving Species One at a Time

- Methods to preserve individual species often involve keeping and breeding the species in captivity.

Captive-Breeding Programs

- Wildlife experts may attempt to restore the population of a species through captive breeding programs.
- These programs involve breeding species in captivity, with the hopes of reintroducing populations to their natural habitats.
- This type of program has been used successfully with the California condor, for example. But the question remains whether or not these restored populations will ever reproduce in the wild.

Preserving Genetic Material

- germ plasma is hereditary material (chromosomes and genes) that is usually contained in the protoplasm of germ cells and may be stored as seeds, sperm, eggs, or pure DNA.

More Study Needed

- Also, small populations are vulnerable to infectious diseases and genetic disorders caused by inbreeding.

Preserving Habitats and Ecosystems

- The most effective way to save species is to protect their habitats.
- Therefore, protecting the habitats of endangered and threatened species often means preserving large areas.

Conservation Strategies

- One strategy is to establish corridors areas of native habitat that can be preserved, restored, and linked into large networks.
- Another promising strategy is to buy products that have been harvested with sustainable practices.

Legal Protection for Species

- The endangered species act is designed to protect any plant or animal species in danger of extinction.

- Under the first provision of the Endangered Species Act, the U.S. Fish and Wildlife Service (USFWS) must compile a list of all endangered and threatened species in the United States.
- As of 2002, 1264 species of plants and animals were listed.
- The second main provision of the act protects listed species from human harm.
- The third provision prohibits the federal government from carrying out any project that jeopardizes a listed species.

Recovery Plans

- Under the fourth main provision of the Endangered Species Act, the USFWS must develop a species recovery plan for each listed species.

Habitat Conservation Plans

- A habitat conservation plan is a land-use plan that attempts to protect threatened or endangered species across a given area by allowing some tradeoffs between harm to the species and additional conservation commitments among cooperating parties.

International Cooperation

- At the global level, the International Union for the Conservation of Nature and Natural Resources facilitates facilitates efforts to protect species and habitats.
- The IUCN publishes red list of species in danger of extinction around the world, advises governments on ways to manage their natural resources, and works with groups like the World Wildlife Fund to sponsor projects such as attempting to stop poaching in Uganda.

International Trade and Poaching

- One product of the IUCN has been an international treaty called CITES (the Convention on International Trade in Endangered Species).
- The CITES treaty was the first effective effort to stop the slaughter of African elephants being killed by poachers who would then sell the ivory tusks.

The Biodiversity Treaty

- One of the most ambitious efforts to tackle environmental issues on a worldwide scale was the United Nations Conference on Environment and Development, also known as the first earth summit. An important result of the Earth Summit was the Biodiversity Treaty.
- The biodiversity treaty is an international agreement aimed at strengthening national control and preservation of biological resources.
- The treaty's goals are to preserve biodiversity and ensure the sustainable and fair use of genetic resources in all countries.